Data Mining Notes

Chapter 1: Introduction

- Data mining is a technology that blends traditional data analysis methods with sophisticated algorithms for processing large volumes of data. It has also opened up exciting opportunities for exploring and analyzing new types of data and for analyzing old types of data in new ways.
- **Business** Point-of-sale data collection (bar code scanners, radio frequency identification (RFID), and smart card technology) have allowed retailers to collect up-to-the-minute data about customer purchases at the checkout counters of their stores.
- Data mining techniques can be used to support a wide range of business intelligence applications such as **customer profiling, targeted marketing, workflow management, store layout, and fraud detection**. It can also help retailers answer important business questions such as "**Who are the most profitable customers?**" "**What products can be cross-sold or up-sold?**" and "**What is the revenue outlook of the company for next year?"**
- **Medicine, Science, and Engineering** Researchers in medicine, science, and engineering are rapidly accumulating data that is key to important new discoveries.

  Techniques developed in data mining can aid Earth scientists in answering questions such as "**What is the relationship between the frequency and intensity of ecosystem disturbances such as droughts and hurricanes to global warming?**" "**How is land surface precipitation and temperature affected by ocean surface temperature?**" and "**How well can we predict the beginning and end of the growing season for a region?**"

What is Data Mining ?

- **Data mining** is the process of automatically discovering useful information in large data repositories. Data mining techniques are deployed to scour large databases in order to find novel and useful patterns that might otherwise remain unknown.
- For example, looking up individual records using a database management system or finding particular Web pages via a query to an Internet search engine are tasks related to the area of **information retrieval**.

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Data Mining and Knowledge Discovery

- Data mining is an integral part of knowledge discovery in databases (KDD), which is the overall process of converting raw data into useful information, as shown in Figure 1.1. This process consists of a series of transformation steps, from data preprocessing to postprocessing of data mining results.

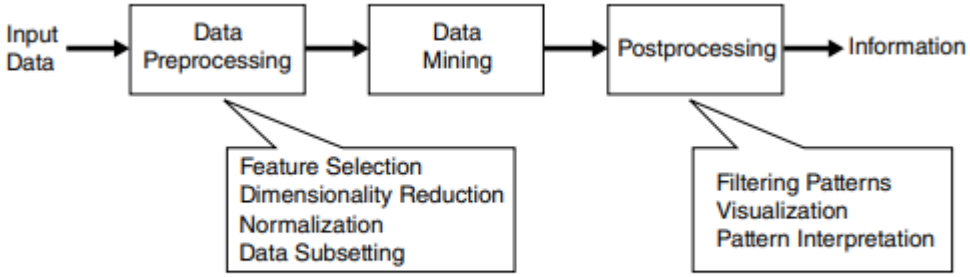

Figure 1.1. The process of knowledge discovery in databases (KDD).

- The **input data** can be stored in a variety of formats (flat files, spread-sheets, or relational tables) and may reside in a centralized data repository or be distributed across multiple sites.
- The purpose of **preprocessing** is to transform the raw input data into an appropriate format for subsequent analysis. The steps involved in data **preprocessing** include fusing data from multiple sources, cleaning data to remove noise and duplicate observations, and selecting records and features that are relevant to the data mining task at hand.
- "**Closing the loop**" is the phrase often used to refer to the process of integrating data mining results into decision support systems.
- In business applications, the insights offered by data mining results can be integrated with campaign management tools so that effective marketing promotions can be conducted and tested. Such integration requires a **postprocessing** step that ensures that only valid and useful results are incorporated into the decision support system.
- Statistical measures or hypothesis testing methods can also be applied during postprocessing to eliminate spurious data mining results.

Motivating Challenge

- As mentioned earlier, **traditional data analysis** techniques have often encountered practical difficulties in meeting the challenges posed by new data sets. The following are some of the specific **challenges** that motivated the development of data mining.

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Scalability

- Because of advances in data generation and collection, data sets with sizes of gigabytes, terabytes, or even petabytes are becoming common. If data mining algorithms are to handle these massive data sets, then they must be scalable.
- Many data mining algorithms employ special search strategies to handle exponential search problems.
- Scalability may also require the implementation of novel data structures to access individual records in an efficient manner. Scalability can also be improved by using sampling or developing parallel and distributed algorithms.

High Dimensionality

- It is now common to encounter data sets with hundreds or thousands of attributes instead of the handful common a few decades ago.
- In bioinformatics, progress in microarray technology has produced gene expression data involving thousands of features.
- Data sets with temporal or spatial components also tend to have high dimensionality.
- For example, consider a data set that contains measurements of temperature at various locations. If the temperature measurements are taken repeatedly for an extended period, the number of dimensions (features) increases in proportion to Motivating Challenges the number of measurements taken.
- Traditional data analysis techniques that were developed for low-dimensional data often do not work well for such high-dimensional data. Also, for some data analysis algorithms, the computational complexity increases rapidly as the dimensionality (the number of features) increases.

Heterogeneous and Complex Data

- Traditional data analysis methods often deal with data sets containing attributes of the same type, either continuous or categorical.
- Examples of such non-traditional types of data include collections of Web pages containing semi-structured text and hyperlinks; DNA data with sequential and three-dimensional structure; and climate data that consists of time series measurements (temperature, pressure, etc.) at various locations on the Earth's surface.

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- Techniques developed for mining such complex objects should take into consideration relationships in the data, such as temporal and spatial autocorrelation, graph connectivity, and parent-child relationships between the elements in semi-structured text and XML documents.

Data Ownership and Distribution

- Sometimes, the data needed for an analysis is not stored in one location or owned by one organization. Instead, the data is geographically distributed among resources belonging to multiple entities. This requires the development of distributed data mining techniques. Among the key challenges faced by distributed data mining algorithms include
  (1) How to reduce the amount of communication needed to perform the distributed computation,
  (2) How to effectively consolidate the data mining results obtained from multiple sources, and
  (3) How to address data security issues.

Non-traditional Analysis

- The traditional statistical approach is based on a hypothesize-and-test paradigm. In other words, a hypothesis is proposed, an experiment is designed to gather the data, and then the data is analyzed with respect to the hypothesis. Unfortunately, this process is extremely labor-intensive.
- Current data analysis tasks often require the generation and evaluation of thousands of hypotheses, and consequently, the development of some data mining techniques has been motivated by the desire to automate the process of hypothesis generation and evaluation.

The Origins of Data Mining

- In particular, data mining draws upon ideas, such as
  (1) Sampling, estimation, and hypothesis testing from statistics and
  (2) search algorithms, modeling techniques, and learning theories from artificial intelligence, pattern recognition, and machine learning.
- Data mining has also been quick to adopt ideas from other areas, including optimization, evolutionary computing, information theory, signal processing, visualization, and information retrieval.

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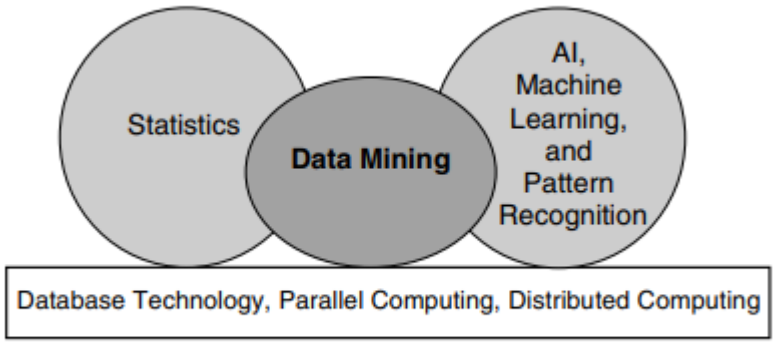

Figure 1.2. Data mining as a confluence of many disciplines.

- A number of other areas also play key supporting roles. In particular, database systems are needed to provide support for efficient storage, indexing, and query processing. Techniques from high performance (parallel) computing are often important in addressing the massive size of some data sets. Distributed techniques can also help address the issue of size and are essential when the data cannot be gathered in one location.

Data Mining Tasks

- Data mining tasks are generally divided into two major categories:

Predictive tasks

- The objective of these tasks is to predict the value of a particular attribute based on the values of other attributes. The attribute to be predicted is commonly known as the **target or dependent variable**, while the attributes used for making the prediction are known as the **explanatory** or **independent variables**.

Descriptive tasks

- The objective is to **derive patterns (correlations, trends, clusters, trajectories, and anomalies)** that summarize the underlying relationships in data. Descriptive data mining tasks are often exploratory in nature and frequently require postprocessing techniques to validate and explain the results.

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Figure 1.3 illustrates four of the core data mining tasks that are described in the remainder of this book.

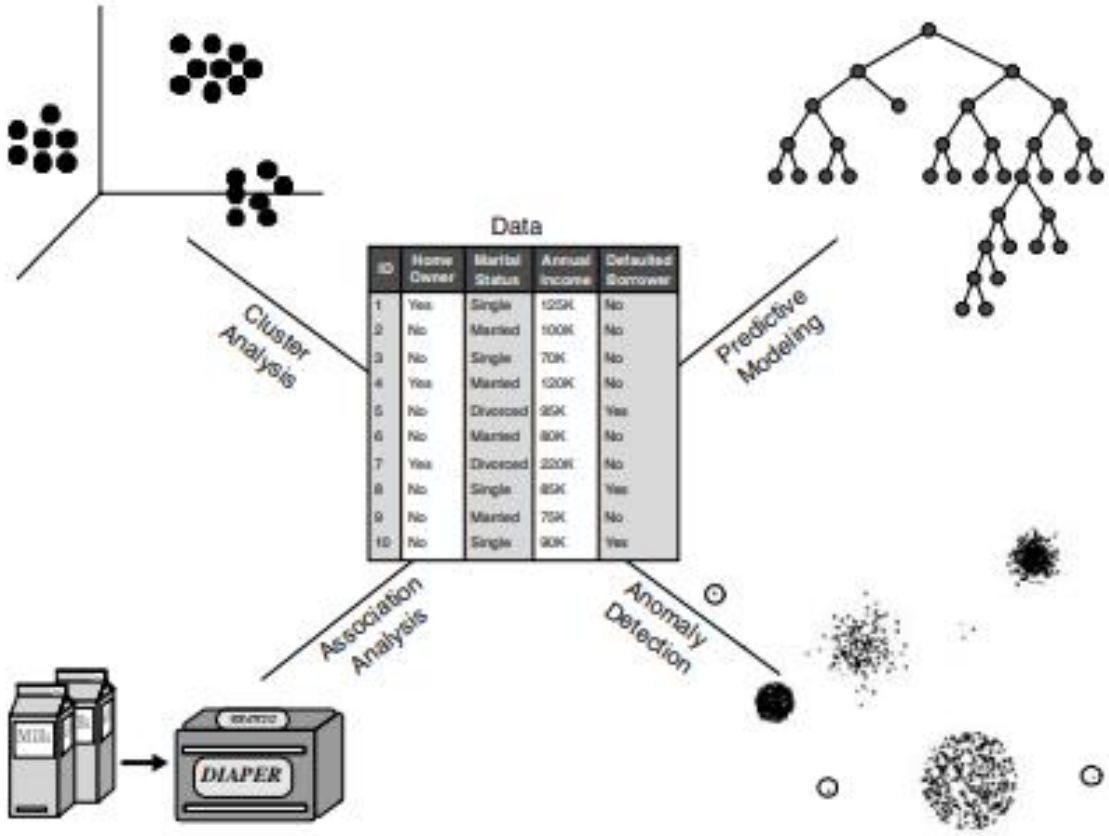

Figure 1.3. Four of the core data mining tasks.

Predictive modeling

- **Predictive** refers to the task of building a model for the target variable as a function of the explanatory variables. There are two types of predictive modeling tasks:
  - **classification**, which is used for discrete target variables, and
  - **regression**, which is used for continuous target variables.

Example 1.1 (Predicting the Type of a Flower)

- Consider the task of predicting a species of flower based on the characteristics of the flower. In particular, consider classifying an Iris flower as to whether it belongs to one of the following three Iris species: Setosa, Versicolour, or Virginica. To perform this task, we need a data set containing the characteristics of various flowers of these three species. A data set with this type of information is the

well-known Iris data set from the UCI Machine Learning Repository at http://www.ics.uci.edu/~mlearn.

- In addition to the species of a flower, this data set contains four other attributes: sepal width, sepal length, petal length, and petal width. (The Iris data set and its attributes are described further in Section 3.1.) Figure 1.4 shows a plot of petal width versus petal length for the 150 flowers in the Iris data set. Petal width is broken into the categories low, medium, and high, which correspond to the intervals [0, 0.75), [0.75, 1.75), [1.75, ∞), respectively. Also, petal length is broken into categories low, medium, and high, which correspond to the intervals [0, 2.5), [2.5, 5), [5, ∞), respectively. Based on these categories of petal width and length, the following rules can be derived:

  Petal width low and petal length low implies Setosa.
  Petal width medium and petal length medium implies Versicolour.
  Petal width high and petal length high implies Virginica.

  While these rules do not classify all the flowers, they do a good (but not perfect) job of classifying most of the flowers. Note that flowers from the Setosa species are well separated from the Versicolour and Virginica species with respect to petal width and length, but the latter two species overlap somewhat with respect to these attributes.

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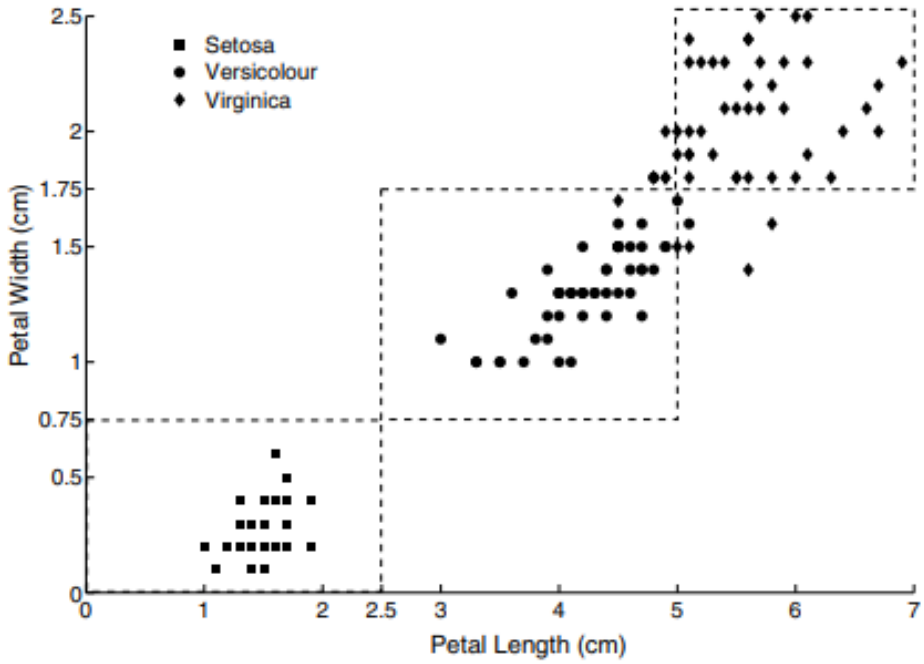

Figure 1.4. Petal width versus petal length for 150 Iris flowers.

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Association analysis

- **Association analysis** is used to discover patterns that describe strongly associated features in the data. The discovered patterns are typically represented in the form of implication rules or feature subsets. Because of the exponential size of its search space, the goal of association analysis is to extract the most interesting patterns in an efficient manner. Useful applications of association analysis include finding groups of genes that have related functionality, identifying Web pages that are accessed together, or understanding the relationships between different elements of Earth's climate system.

Example 1.2 (Market Basket Analysis)

- The transactions shown in Table 1.1 illustrate point-of-sale data collected at the checkout counters of a grocery store. Association analysis can be applied to find items that are frequently bought together by customers. For example, we may discover the rule {**Diapers**} → {**Milk**}, which suggests that customers who buy diapers also tend to buy milk. This type of rule can be used to identify potential cross-selling opportunities among related items.

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Table 1.1. Market basket data.

Transaction ID	Items
1	{Bread, Butter, Diapers, Milk}
2	{Coffee, Sugar, Cookies, Salmon}
3	{Bread, Butter, Coffee, Diapers, Milk, Eggs}
4	{Bread, Butter, Salmon, Chicken}
5	{Eggs, Bread, Butter}
6	{Salmon, Diapers, Milk}
7	{Bread, Tea, Sugar, Eggs}
8	{Coffee, Sugar, Chicken, Eggs}
9	{Bread, Diapers, Milk, Salt}
10	{Tea, Eggs, Cookies, Diapers, Milk}

Cluster analysis

- **Cluster analysis** seeks to find groups of closely related observations so that observations that belong to the same cluster are more similar to each other than observations that belong to other clusters. Clustering has been used to group sets of related customers, find areas of the ocean that have a significant impact on the Earth's climate, and compress data.

Example 1.3 (Document Clustering)

- The collection of news articles shown in Table 1.2 can be grouped based on their respective topics. Each article is represented as a set of word-frequency pairs (w, c), where w is a word and c is the number of times the word appears in the article. There are two natural clusters in the data set. The first cluster consists of the first four articles, which correspond to news about the economy, while the second cluster contains the last four articles, which correspond to news about health care. A good clustering algorithm should be able to identify these two clusters based on the similarity between words that appear in the articles.

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Table 1.2. Collection of news articles.

Article	Words
1	dollar: 1, industry: 4, country: 2, loan: 3, deal: 2, government: 2
2	machinery: 2, labor: 3, market: 4, industry: 2, work: 3, country: 1
3	job: 5, inflation: 3, rise: 2, jobless: 2, market: 3, country: 2, index: 3
4	domestic: 3, forecast: 2, gain: 1, market: 2, sale: 3, price: 2
5	patient: 4, symptom: 2, drug: 3, health: 2, clinic: 2, doctor: 2
6	pharmaceutical: 2, company: 3, drug: 2, vaccine: 1, flu: 3
7	death: 2, cancer: 4, drug: 3, public: 4, health: 3, director: 2
8	medical: 2, cost: 3, increase: 2, patient: 2, health: 3, care: 1

Anomaly detection

- **Anomaly detection** is the task of identifying observations whose characteristics are significantly different from the rest of the data. Such observations are known as **anomalies** or **outliers**. The goal of an anomaly detection agorithm is to discover the real anomalies and avoid falsely labeling normal objects as anomalous. In other words, a good anomaly detector must have a high detection rate and a low false alarm rate. Applications of anomaly detection include the detection of fraud, network intrusions, unusual patterns of disease, and ecosystem disturbances.

Example 1.4 (Credit Card Fraud Detection)

- A credit card company records the transactions made by every credit card holder, along with personal information such as credit limit, age, annual income, and address. Since the number of fraudulent cases is relatively small compared to the number of legitimate transactions, anomaly detection techniques can be applied to build a profile of legitimate transactions for the users. When a new transaction arrives, it is compared against the profile of the user. If the characteristics of the transaction are very different from the previously created profile, then the transaction is flagged as potentially fraudulent.

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Data Mining Notes

Chapter 2: Data

The Type of Data

- Data sets differ in a number of ways. For example, the attributes used to describe data objects can be of different types—**quantitative** or **qualitative**—and data sets may have special characteristics; e.g., some data sets contain time series or objects with explicit relationships to one another. Not surprisingly, the type of data determines which tools and techniques can be used to analyze the data. Furthermore, new research in data mining is often driven by the need to accommodate new application areas and their new types of data.

The Quality of the Data

- Data is often far from perfect. While most data mining techniques can tolerate some level of imperfection in the data, a focus on understanding and improving data quality typically improves the quality of the resulting analysis. **Data quality issues** that often need to be addressed include the **presence of noise and outliers; missing, inconsistent, or duplicate data; and data that is biased** or, in some other way, unrepresentative of the phenomenon or population that the data is supposed to describe.

Pre-processing Steps to Make the Data More Suitable for Data Mining

- Often, the raw data must be processed in order to make it suitable for analysis. While one objective may be to improve data quality, other goals focus on modifying the data so that it better fits a specified **data mining technique or tool**. For example, a continuous attribute, e.g., length, may need to be transformed into an attribute with discrete categories, e.g., short, medium, or long, in order to apply a particular technique.

Analyzing Data in Terms of Its Relationships

- One approach to data analysis is to find **relationships** among the data objects and then perform the remaining analysis using these relationships rather than the data objects themselves. For instance, we can compute the similarity or distance between pairs of objects and then perform the analysis—**clustering, classification, or anomaly detection** based on these similarities or distances. There are many such similarity or distance

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measures, and the proper choice depends on the type of data and the particular application.

Types of Data

A **data** set can often be viewed as a combined collection of **data objects**. Other names for a data object are **record**, **point**, **vector**, **pattern**, **event**, **case**, **sample**, **observation**, or **entity**.

In turn, data objects are described by a number of attributes that capture the basic characteristics of an object, such as the mass of a physical object or the time at which an event occurred. Other names for an attribute are **variable**, **characteristic**, **field**, **feature**, or **dimension.**

Attributes and Measurement

Attribute :- An **attribute** is a property or characteristic of an object that may vary, either from one object to another or from one time to another.

> **For example**, eye color varies from person to person, while the temperature of an object varies over time. Note that eye color is a symbolic attribute with a small number of possible values {brown, black, blue, green, hazel, etc.}, while temperature is a numerical attribute with a potentially unlimited number of values.

Measurement :- The process of **measurement** is the application of a measurement scale to associate a value with a particular attribute of a specific object.

Measurement Scale :- A measurement scale is a rule (function) that associates a numerical or symbolic value with an attribute of an object.

The Different Types of Attributes

The following properties (operations) of numbers are typically used to describe attributes.

1. Distinctness; = and !=

2. Order; <, ≤, >, and ≥

3. Addition; + and −

4. Multiplication; ∗ and /

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Table 2.2. Different attribute types.

Attribute Type		Description	Examples	Operations
Categorical (Qualitative)	Nominal	The values of a nominal attribute are just different names; i.e., nominal values provide only enough information to distinguish one object from another. $(=, \neq)$	zip codes, employee ID numbers, eye color, gender	mode, entropy, contingency correlation, χ^2 test
	Ordinal	The values of an ordinal attribute provide enough information to order objects. $(<, >)$	hardness of minerals, $\{good, better, best\}$, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Numeric (Quantitative)	Interval	For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists. $(+, -)$	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviatio Pearson's correlation, t and F tests
	Ratio	For ratio variables, both differences and ratios are meaningful. $(*, /)$	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current	geometric mean, harmonic mean, percent variation

Table 2.3. Transformations that define attribute levels.

Attribute Type		Transformation	Comment
Categorical (Qualitative)	Nominal	Any one-to-one mapping, e.g., a permutation of values	If all employee ID numbers are reassigned, it will not make any difference.
	Ordinal	An order-preserving change of values, i.e., $new_value = f(old_value)$, where f is a monotonic function.	An attribute encompassing the notion of good, better, best can be represented equally well by the values $\{1, 2, 3\}$ or by $\{0.5, 1, 10\}$.
Numeric (Quantitative)	Interval	$new_value = a * old_value + b$, a and b constants.	The Fahrenheit and Celsius temperature scales differ in the location of their zero value and the size of a degree (unit).
	Ratio	$new_value = a * old_value$	Length can be measured in meters or feet.

Describing Attributes by the Number of Values

Discrete :- A **discrete** attribute has a finite or countably infinite set of values. Such attributes can be **categorical**, such as zip codes or ID numbers, or numeric, such as counts. **Discrete** attributes are often represented using integer variables. **Binary** attributes are a special case of discrete attributes and assume only two values, e.g., true/false, yes/no, male/female, or 0/1. **Binary** attributes

are often represented as **Boolean** variables, or as **integer** variables that only take the values 0 or 1.

Continuous :- A **continuous** attribute is one whose values are real numbers. **Examples** include attributes such as temperature, height, or weight. Continuous attributes are typically represented as floating-point variables. Practically, real values can only be measured and represented with limited precision.

Typically, **nominal** and **ordinal** attributes are **binary** or **discrete**, while **interval** and **ratio** attributes are **continuous**. However, **count** attributes, which are **discrete**, are also **ratio** attributes.

Asymmetric Attributes

For **asymmetric** attributes, only presence a non-zero attribute value is regarded as important.

Asymmetric binary attributes :- Binary attributes where only non-zero values are important are called **asymmetric binary attributes**.

Asymmetric discrete or **continuous attributes** :- If the number of credits associated with each course is recorded, then the resulting data set will consist of **asymmetric discrete** or **continuous attributes**.

Types of Data Sets

For convenience, we have grouped the types of data sets into three groups: **record data, graph-based data,** and **ordered data**. These categories do not cover all possibilities and other groupings are certainly possible.

General Characteristics of Data Sets

Dimensionality :- The **dimensionality** of a data set is the number of attributes that the objects in the data set possess. Data with a small number of dimensions tends to be qualitatively different than moderate or high-dimensional data. Indeed, the difficulties associated with analyzing high-dimensional data are sometimes referred to as the **curse of dimensionality**. Because of this, an important motivation in preprocessing the data is **dimensionality reduction**.

Sparsity :- For some data sets, such as those with asymmetric features, most attributes of an object have values of 0; in many cases, fewer than 1% of the entries are non-zero. In practical terms, **sparsity** is an advantage because usually only the non-zero values need to be stored and manipulated. This results in significant savings with respect to computation time and storage. Furthermore, some data mining algorithms work well only for sparse data.

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Resolution :- It is frequently possible to obtain data at different levels of resolution, and often the properties of the data are different at different resolutions. For instance, the surface of the Earth seems very uneven at a resolution of a few meters, but is relatively smooth at a resolution of tens of kilometers. The patterns in the data also depend on the level of resolution. If the resolution is too fine, a pattern may not be visible or may be buried in noise; if the resolution is too coarse, the pattern may disappear. For example, variations in atmospheric pressure on a scale of hours reflect the movement of storms and other weather systems. On a scale of months, such phenomena are not detectable.

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